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Convergent representation of values from tactile and visual inputs for efficient goal-directed behavior in the primate putamen

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Animals can discriminate diverse sensory values with a limited number of neurons, raising questions about how the brain utilizes neural resources to efficiently process multi-dimensional inputs for decision-making. Here, we demonstrate that this efficiency is achieved by reducing sensory dimensions and converging towards the value dimension essential for goal-directed behavior in the putamen. Humans and monkeys performed tactile and visual value discrimination tasks while their neural responses were examined. Value information, whether originating from tactile or visual stimuli, was found to be processed within the human putamen using fMRI. Notably, at the single-neuron level in the macaque putamen, half of the individual neurons encode values independently of sensory inputs, while the other half selectively encode tactile or visual value. The responses of bimodal value neurons correlate with value-guided finger insertion behavior in both tasks, whereas modality-selective value neurons show task-specific correlations. Simulation using these neurons reveals that the presence of bimodal value neurons enables value discrimination with a significantly reduced number of neurons compared to simulations without them. Our data indicate that individual neurons in the primate putamen process different values in a convergent manner, thereby facilitating the efficient use of constrained neural resources for value-guided behavior.

Various types of values can originate from sensory inputs such as tactile and visual sensations. The basal ganglia (BG) system might process various values from these diverse sensory inputs because neurons in the cortical regions that process each type of sensory information innervate structures in the BG[1–3].

Notably, the number of neurons is known to decrease as information flows from the cortex to each successive structure in the BG[4–6]. This anatomical observation raises the question of how the brain processes such diverse sensory values with a decreasing number of neurons. One possibility for reducing the number of neurons involved in processing of various values is that neurons extract value information directly related with a goal, regardless of diverse sensory inputs. Indeed, the convergent information process was proposed solely based on the anatomical funneling structure, where the number of neurons decreased through each successive structure of the basal ganglia[7,8]. However, the lack of empirical data supporting this

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convergent process leaves another possibility that each sensory value is processed independently to retain both value and sensory modality information, even with a limited number of neurons. This question could be explored through hypotheses that persist in an ongoing debate and remain unresolved: parallel information processing and convergent information processing in the basal ganglia circuits, as suggested previously[4–6,8–10]. Investigating whether a convergent value process exists in the BG system is crucial for understanding how our brain efficiently utilizes a limited number of neurons for cognitive function.

Primates can discriminate diverse values not only by using their eyes but also by using their hands and fingers for decision-making[11,12]. For example, humans can find a dropped coin among various objects beneath a sofa solely through tactile perception using their fingers. Similar searching behavior is commonly observed among primates as they skillfully use their finger perception to sift through and select valuable objects placed inside various cavities or crevices[13–16]. This foraging behavior with finger exploration provides insights into the cognitive processes through which primates discriminate tactile stimuli, remember their values, and make decisions based on their tactile perception through fingertips. However, there has been little research on the brain regions responsible for processing tactile value information, with most studies focusing on cognitive processes related to visual values in the primate brain.

Previous studies on visual value processing have shown that structures in the BG system, including the putamen and caudate, process values of visual stimuli, providing insights into the tactile value process[17–20]. Within the BG structures, the putamen is known to anatomically receive inputs from the somatosensory cortex, including finger regions, as well as from the temporal cortex for visual processing[3,21,22]. Furthermore, midbrain dopamine neurons, crucial for value learning through the reward prediction error signal, strongly innervate the putamen[23,24]. These anatomical connections suggest a role for the putamen in processing both visual and tactile values.

If tactile and visual values are processed in the putamen, we can test whether they converge or are processed in parallel by investigating the value-coding responses in single neurons of the putamen. In the case of convergent value processing, our finding will indicate that a single neuron encodes both values, regardless of sensory inputs. Conversely, if tactile and visual values are processed independently, we will observe that a single neuron encodes either tactile or visual value. Thus, examining the neural encoding of tactile and visual values in the primate putamen is critical to elucidate whether the BG system utilizes convergent representation of value for efficient goal-directed behavior with a limited number of neurons.

In this study, we screened the striatal regions using functional magnetic resonance imaging (fMRI) and discovered that the human putamen represented both tactile and visual value information. To reveal how individual neurons process these different sensory values, we examined the responses of single neurons to each object in the macaque putamen while monkeys performed both tactile and visual value discrimination tasks. We then conducted a simulation using the modality-selective value neurons and bimodal value neurons identified through our single-unit recording to investigate how the putamen neurons efficiently process diverse sensory values with constrained neural resources.

Results

Representation of both tactile and visual values in the human putamen

To identify which striatal structures are involved in processing tactile and visual values, we conducted a human fMRI experiment with two value reversal tasks: the tactile task and the visual task (Fig. 1). Participants experienced both tactile (braille) and visual stimuli (fractal image) in each trial of the value reversal tasks (Fig. 1a). In the tactile task, gain or loss of a determined amount of monetary reward was associated exclusively with the identity of the tactile stimulus, and in the visual task, it was associated only with the identity of the visual stimulus (Fig. 1b). This association between stimuli and monetary rewards generated deterministic values of stimuli as previously described[25]. After experiencing these stimuli, participants were instructed to press a button to indicate whether the stimulus was associated with a good or bad monetary reward, and then they received feedback about their response (Fig. 1c). To distinguish brain responses for value processing from those for stimulus identity processing, the stimulus-reward contingency was reversed in the middle of each task run (Fig. 1b). We found that participants successfully learned the stimulus-value associations and made responses aimed at maximizing their total monetary gain in both the tactile and visual tasks (tactile task: 96.59 ± 0.59% accuracy; visual task: 96.95 ± 0.64% accuracy; excluding the first trial and the rule-switching trial in each run).

There were no significant differences in accuracy and reaction time (RT) between the two tasks (two-tailed paired *t*-test, accuracy: $t_{(21)} = -0.458$, $p = 0.652$; RT: $t_{(21)} = -1.123$, $p = 0.274$). Moreover, accuracy for the trials immediately following reversal, reflecting the learning rate of value information, also showed comparable results between the two tasks (two-tailed paired *t*-test, $t_{(21)} = -1.227$, $p = 0.234$). Additionally, there was no difference in the duration of stimuli presentation-guided visually but subjectively adjusted based on necessary recognition time-between the tasks (two-tailed paired *t*-test, $t_{(21)} = 1.157$, $p = 0.260$). These behavioral results suggest the successful acquisition of value information across both sensory modalities, with minimal task-related differences.

Next, we examined which striatal structures represented each type of value information based on the human BOLD responses. Using a searchlight procedure, we identified structures with positive value discrimination indices. These indices were computed based on the mean voxel pattern correlation between trials with the same value (e.g., good-and-good or bad-and-bad) subtracted by the mean correlation between trials with opposite values (good-and-bad) (Fig. 1d, left panel)[26,27]. Notably, our searchlight analysis revealed a significant value discrimination response in the putamen of the left hemisphere during the tactile task, corresponding to the side contralateral to tactile perception by the right fingertips (Fig. 1d, middle panel). During the visual task, we found a significant value discrimination response in the bilateral putamen, as well as in the caudate and right ventral striatum (Fig. 1d, right panel). These results demonstrate that the human putamen represented both tactile and visual values, implying the potential for value-convergent processing in the human putamen.

Learning the values of tactile and visual stimuli in macaque monkeys

The human fMRI study revealed that both tactile and visual values were represented in the putamen, raising the question of whether a single neuron converges both values within this structure. To explore how single neurons in the putamen process tactile and visual value information, we designed a tactile value reversal task (T-VRT) and visual value reversal task (V-VRT) using a newly developed tactile stimulus presenter for macaque monkeys (Figs. 2a–d and S1). The two monkeys (UL and EV) were required to insert their fingers into the finger hole to experience a braille pattern as a tactile stimulus in the T-VRT (Figs. 2a and S1b, c) (Movie S1). Similarly, a fractal image was presented on the screen when the monkeys inserted their fingers into the hole in the V-VRT. In each block of these tasks, one tactile or visual stimulus was associated with a reward (good stimulus), and the other stimulus was not (bad stimulus) (Fig. 2b). This stimulus-liquid reward association generated a deterministic value for stimulus, based on the consistent amount and presence of a reward in a block[28–30]. This association was reversed in each block to investigate neural responses encoding values while excluding neural responses to the stimuli themselves.

To evaluate whether monkeys learned the tactile and visual values in these tasks, we first examined the differences in finger insertion reaction times (RTs) when they anticipated a good or bad stimulus. Each stimulus was presented twice within a single-stimulus presentation trial (Figs. 2c, d, upper schemes and S2a–c). If the monkeys recognized the value during the initial stimulus presentation and anticipated it before the second finger insertion, this guided their value-based finger insertion behavior. In an example session, the monkey demonstrated quicker finger insertions in response to the second finger-in cue when anticipating good stimuli than when anticipating bad stimuli in both tasks (two-tailed unpaired *t*-test, $p = 2.646 \times 10^{-6}$ in T-VRT; $p = 2.116 \times 10^{-6}$ in V-VRT) (Fig. 2e).

We analyzed the differences in finger insertion RTs across all sessions for both tasks. In the T-VRT, 84% of sessions (255 out of 299) and in the V-VRT, 91% of sessions (273 out of 299) showed significant differences in finger insertion RTs, with faster finger insertion for anticipated good stimuli than bad stimuli ($p < 0.05$, two-tailed unpaired *t*-test for each neuron) (Figs. 2f and S2d, e). In addition, we found that the monkeys were more inclined to avoid touching the bad stimuli than the good stimuli (paired *t*-test, $p = 5.339 \times 10^{-40}$ for T-VRT, $p = 1.984 \times 10^{-70}$ for V-VRT) (Fig. S2f). The acquisition of tactile and visual values was further confirmed by their preference for good stimuli in choice trials (Fig. 2c, d, lower schemes). The monkeys successfully selected good stimuli more than bad stimuli in both tasks (paired *t*-test, $p = 2.655 \times 10^{-215}$ for T-VRT, $p = 6.758 \times 10^{-312}$ for V-VRT; 92.72 ± 0.48% and 97.66 ± 0.25% average correct choice rate for the T-VRT and V-VRT, respectively) (Figs. 2g and S2g, h) (Movies S2). This value-guided finger insertion behavior indicates that the monkeys successfully acquired values associated with the tactile and visual stimuli, enabling us to investigate the neural encoding of these tactile and visual values.

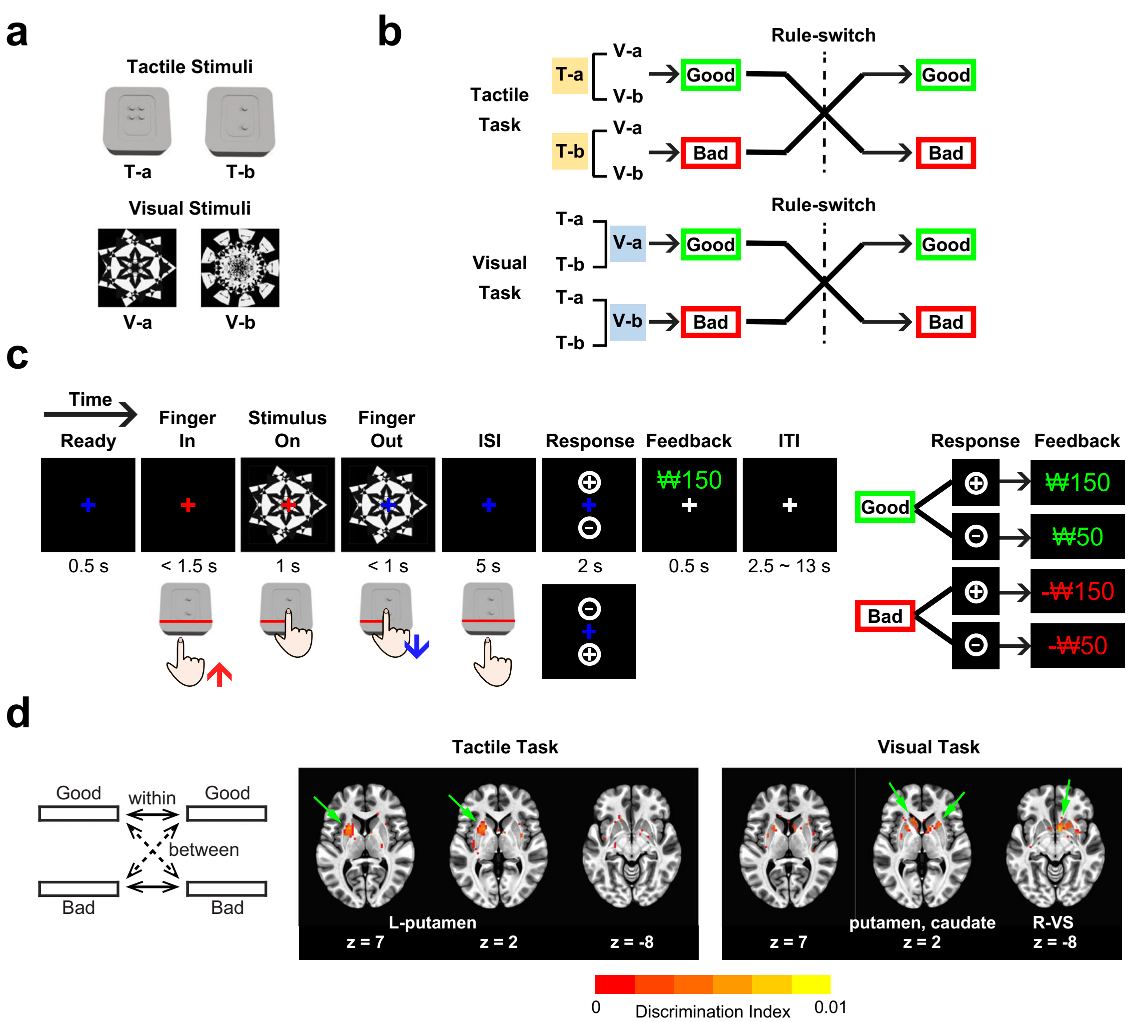

Fig. 1 | Representation of both tactile and visual values in the human putamen. **a** Tactile and visual stimuli for human participants. Two braille patterns and two fractal images were used as the tactile and visual stimuli, respectively. **b** Value reversal scheme. In each task, either tactile or visual stimuli were associated with monetary values, and stimuli from the other modality were devoid of monetary associations. Within each task run, one stimulus was linked to monetary gain, and the other was paired with monetary loss. This stimulus-reward contingency was reversed in the middle of each run of both tasks. **c** Tactile and visual value reversal tasks. Participants were presented with both tactile and visual stimuli simultaneously, with one modality being associated with a monetary gain or loss according to the task. Participants were instructed to press a button indicating whether the stimulus was associated with a good or bad monetary reward. Correct responses to stimuli associated with a monetary gain earned participants 150 won, and incorrect responses provided 50 won (good stimuli). In contrast, correct responses to stimuli associated with a monetary loss deducted 50 won, and incorrect responses deducted 150 won (bad stimuli). **d** Searchlight results showing both tactile and visual value representations in the putamen. The discrimination index for value information was calculated as the difference between the averages of within-value correlations and between-value correlations. The colored areas indicate significant discrimination of value across participants (right-tailed *t*-test; $p < 0.05$, $t_{(21)} > 1.721$) delineated on axial slices of the MNI brain.

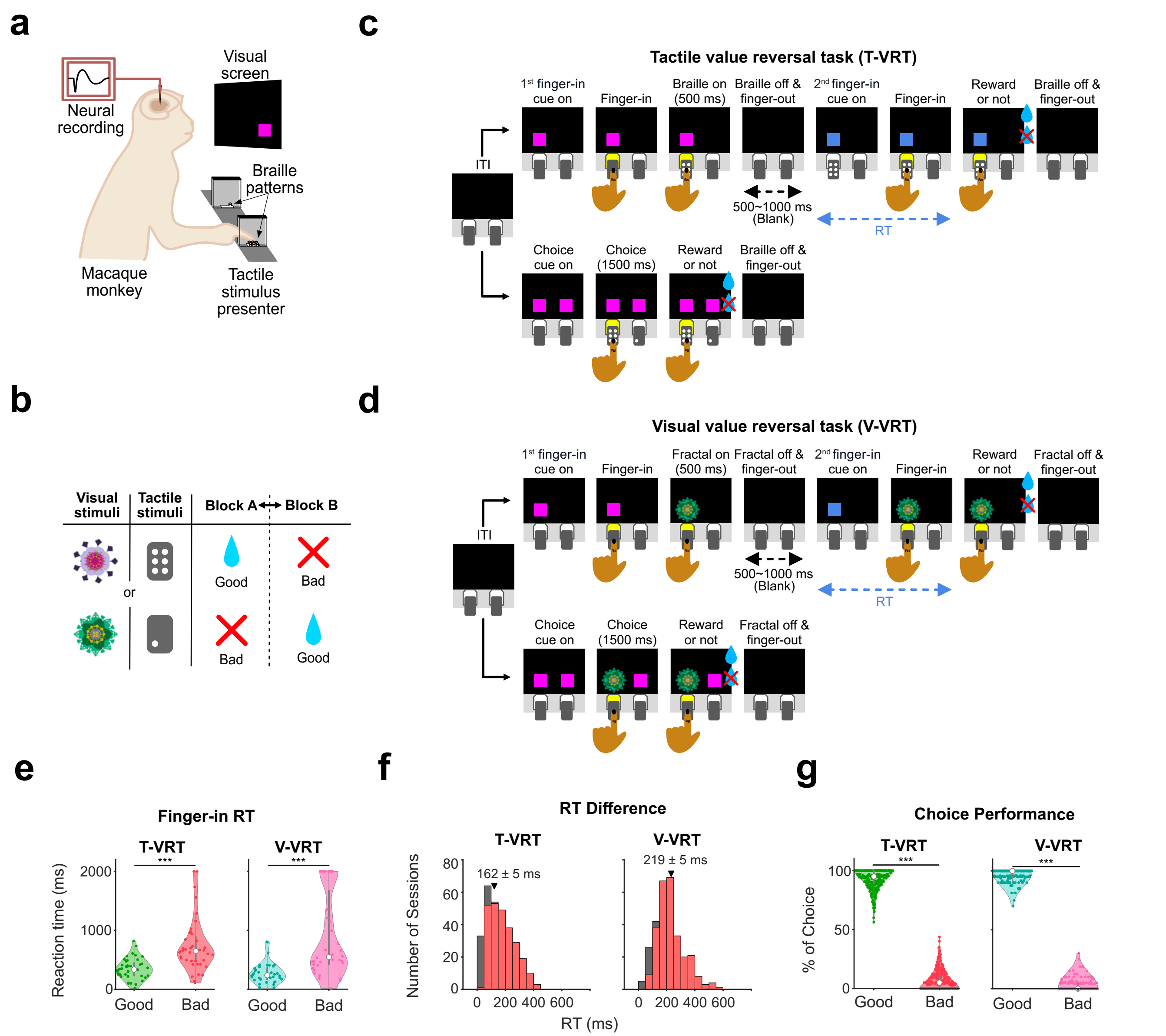

Fig. 2 | Tactile and visual value learning in macaque monkeys. a Schematic of the experimental conditions. Tactile and visual stimuli were presented through the braille presenter and monitor screen, respectively. Each monkey's head was fixed for electrophysiology and viewing visual stimuli on the screen. **b** Reversal of value for tactile and visual stimuli. In each block, either a braille or fractal stimulus was associated with a liquid reward, and the other was not. This stimulus-reward contingency was then reversed in the following block. **c** Tactile value reversal task (T-VRT). The T-VRT includes two types of trials: single-stimulus trials and choice trials. In the single-stimulus trials, the monkeys experienced a single tactile stimulus and received the associated reward. In the choice trials, two squares for selection were presented. **d** Visual value reversal task (V-VRT). The procedure for the V-VRT was identical to that for the T-VRT except for the modality of the stimuli. **e** Example of faster reaction times for finger insertion for good stimuli than bad stimuli. The monkeys inserted their fingers more quickly for anticipated good stimuli than bad stimuli in response to the second finger-in cue ($n = 40$ and $n = 36$ for the T-VRT and V-VRT, respectively). The solid thick gray bar within each violin plot represents the inter-quantile range (from the 1st to the 3rd quantile) of the data points. The white circle within each plot indicates the median value. $^{***}p < 0.0005$, two-tailed unpaired t-test was applied. **f** Histogram of averaged differences in finger insertion reaction times measured across all the recording sessions for the two monkeys ($n = 299$ for both the T-VRT and V-VRT). Red bars indicate sessions with statistically significant differences (two-tailed unpaired t-test, $p < 0.05$). Triangles and numerical values indicate the mean differences from all sessions (mean ± SE). **g** Percentage of choices in the choice trials. The percentages for selecting good and bad stimuli were plotted ($n = 299$ for both the T-VRT and V-VRT). Two-tailed unpaired t-test was applied. $^{***}p < 0.0005$.

Encoding of both tactile and visual values by individual neurons in the monkey putamen

To examine whether the putamen neurons processed value information, we recorded the spike activity of single neurons in the putamen during tactile and visual value tasks. In the T-VRT, the monkeys were instructed to insert their index fingers into a hole upon the appearance of the first finger-in cue (Fig. 2c, upper scheme). Following the insertion of the finger, one of the tactile stimuli was presented for a duration of 500 ms to examine the neural responses encoding the value associated with each braille pattern (Figs. 2c and S2a). In the V-VRT, the monkeys were required to insert their index fingers into the hole to display a fractal image on the screen, ensuring consistent motor movements between both tasks for a fair comparison of the neural responses encoding tactile and visual values (Fig. 2d).

We identified value discrimination responses in the putamen neurons. Figure 3a shows an example neuron in the T-VRT that encoded value information during tactile stimulus presentation, with stronger responses to good tactile stimuli than bad ones (referred to as

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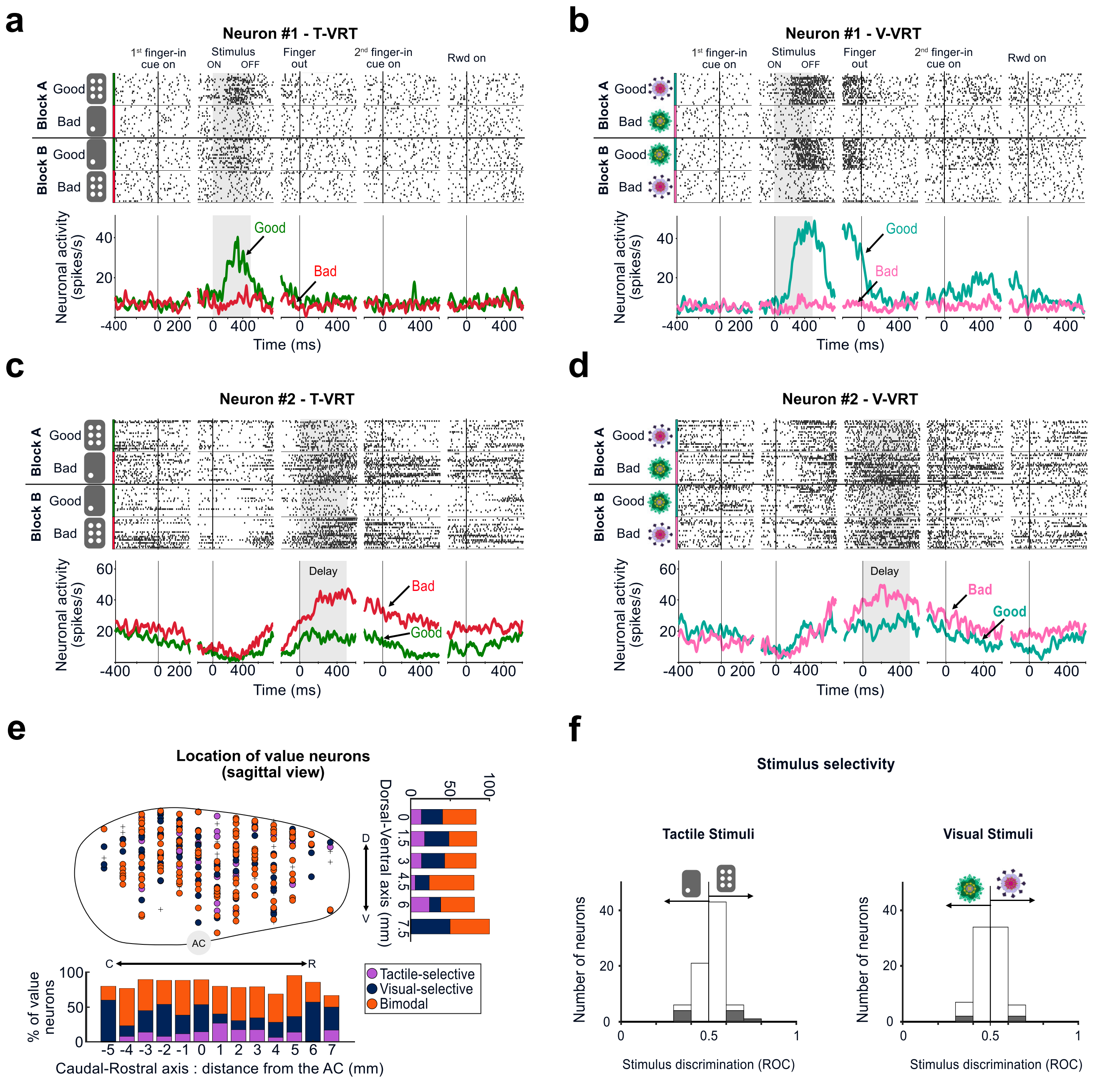

Fig. 3 | Both tactile and visual values encoded in single neurons of the monkey putamen. a, **b** Example of an individual putamen neuron encoding both tactile and visual values during the stimulus presentation period. Neuronal responses to good and bad stimuli are shown in raster plots and peristimulus time histograms (PSTHs). **c**, **d** Example of a single putamen neuron encoding both tactile and visual values during the delay period. Responses of this neuron are in the same format as in (**a**, **b**). **e** Locations of value neurons in the sagittal view. Purple and navy indicate tactile-selective ($n = 41$) and visual-selective ($n = 77$) value neurons, respectively. Orange indicates bimodal value neurons ($n = 129$). Black crosses indicate non-value neurons ($n = 52$). D dorsal, V ventral, R rostral, C caudal, AC anterior commissure. **f** Stimulus discrimination responses of value neurons. For each neuron, the difference in responses to two stimuli was calculated as the ROC area. The gray bars indicate neurons that exhibited statistically significant stimulus selectivity responses in the T-VRT ($n = 77$) and V-VRT ($n = 81$).

a *stimulus value neuron*). Notably, this same neuron also exhibited significantly greater responses to good visual stimuli than bad ones in the V-VRT (Fig. 3b). This example neuron demonstrates that both tactile and visual values were encoded in a single neuron of the putamen.

Furthermore, we discovered another type of neuron that encoded value information during the delay period when the monkeys remained nearly motionless after retracting their fingers (referred to as a *delay value neuron*) (Fig. 3c, d). The example delay value neuron in Fig. 3c exhibited a higher response to bad tactile stimuli than good tactile stimuli during the delay period but not during stimulus presentation in the T-VRT. In the V-VRT, this same neuron also showed value discrimination responses for visual stimuli only during the delay period (Fig. 3d). Our example neural data thus reveal that the putamen contains stimulus and delay value neurons that encode both tactile and visual value information (referred to as *bimodal value neurons*).

Three types of value neurons in the primate putamen

We discovered these bimodal value neurons distributed across the putamen, but we also found modality-selective value neurons that exclusively encoded either tactile or visual values (Figs. 3e and S3a). Among the 299 task-responsive neurons found in the putamen, 247 (82.6%) neurons showed value discrimination responses to both tactile and visual stimuli or to either one. To confirm that these value discrimination activities were not influenced by different finger movements in response to good and bad stimuli, we analyzed the movement patterns during the first finger insertion period (Fig. S4a). Our analysis showed no differences in finger movements when monkeys interacted with stimuli of different values, indicating that the value discrimination responses were not a result of different finger movements (Fig. S4b, c).

The average responses of these three types of value neurons showed distinct value discrimination activities (Figs. 4a–f and S5, S6). Considering that various putamen neurons responded more strongly to either good or bad stimuli (*positive* and *negative value-coding neurons* in Fig. S7), we calculated the average responses based on the neurons' preferred and non-preferred values. The population activities of the stimulus value neurons increased after the presentation of the first finger-in cue, and they displayed distinct responses encoding tactile values, visual values, or both (Fig. 4a–c). *Tactile-selective value neurons* responded more strongly to the preferred tactile value than the non-preferred one, but did not show value discrimination responses to visual stimuli (Fig. 4a). In contrast, *visual-selective value neurons* showed stronger responses to their preferred visual value than the non-preferred one, but they did not exhibit value discrimination activity to tactile stimuli (Fig. 4b). Different from these modality-selective value neurons, bimodal value neurons showed value discrimination responses to both tactile and visual stimuli during stimulus presentation (Fig. 4c).

Similar patterns of value discrimination activity were found in the averaged responses of the delay value neurons after the stimulus disappeared. They showed three types of value-coding responses during the blank delay period after the finger was retracted (Fig. 4d–f).

Electrophysiological features of the value neurons in the putamen

The spike shape and duration of these value neurons were not different from those of the non-value neurons (one-way ANOVA, $F(3, 409) = 2.35$, $p = 0.071$ for spike duration) (Fig. S3b, c). In the comparison of baseline activities, visual-selective value neurons exhibited slightly higher activity levels than tactile-selective and bimodal value neurons, suggesting a potential difference in their input pathways (one-way ANOVA, $F(3,594) = 3.94$, $p = 0.008$; post hoc Bonferroni pairwise comparison, $p = 0.012$ for tactile-selective and visual-selective; $p = 0.037$ for visual-selective and bimodal) (Fig. S3d). The analysis of activity variability for value response revealed no significant difference among the three types of value neurons (one-way ANOVA, $F(3,372) = 2.08$, $p = 0.102$), indicating that value neurons showed similar stability in processing value information. In addition, we compared the onset times of value discrimination responses after the stimulus presentation between the different types of stimulus value neurons (Fig. S8). In both T-VRT and V-VRT, there were no statistical significances in these onset times for value discrimination between modality-selective and bimodal value neurons. (two-tailed unpaired *t*-test, $p = 0.352$ and $p = 0.307$ for T-VRT and V-VRT, respectively).

We next analyzed the stimulus selectivity of these value neurons by comparing their neural responses to pairs of tactile or visual stimuli associated with reward. 11.68% (9/77) of value neurons showed stimulus selectivity to particular tactile stimuli, and 3.93% (4/81) exhibited selectivity to specific visual stimuli during the initial stimulus presentation (two-tailed Wilcoxon rank-sum test, $p < 0.05$) (Fig. 3f). The data suggest that these value neurons, which are widely distributed in the putamen, are primarily involved in processing value information rather than stimulus selectivity.

Half of value neurons converge tactile and visual value information

Interestingly, more than half of the value neurons (52.23%, 129 out of 247) were bimodal value neurons that converged both tactile and visual values in single neurons. These proportions were consistently observed across both monkeys (51.68% and 53.06% in UL and EV, respectively). 53.4% (55 out of 103) of stimulus value neurons and 51.39% (74 out of 144) of delay value neurons were bimodal value neurons, indicating their consistent distribution across these categories of value neurons (Figs. 4g and S3f).

Their value discrimination responses to tactile stimuli showed a strong positive correlation with the discrimination responses to visual stimuli in both bimodal stimulus and delay value neurons ($r = 0.84$, $p = 6.247 \times 10^{-16}$ and $r = 0.85$, $p = 1.148 \times 10^{-22}$, for stimulus and delay neurons, respectively, Pearson correlation coefficient) (Fig. 4h, i). 95.35% of bimodal value neurons encoded tactile and visual values in the same direction, either positive or negative, with only 4.65% of these neurons encoding the values in opposite directions. These results suggest that each bimodal value neuron processes value information in a similar manner in terms of value-encoding direction and strength, regardless of the sensory modality presented.

In the other half of value neurons, 21.36% (22 out of 103) of stimulus value neurons and 13.19% (19 out of 144) of delay value neurons exhibited neural responses selective to tactile value (Fig. 4g). The remainder of these value neurons, 25.24% (26 out of 103) for stimulus neurons and 35.42% (51 out of 144) for delay value neurons, selectively encoded visual values (Fig. 4g). These findings suggest that the putamen processes value information through a combination of parallel and convergent information pathways at the single-neuron level.

The responses of value neurons correlated with anticipated value-guided finger insertion

We have demonstrated that the putamen neurons encoded values independently of movements during the first stimulus presentation (Figs. 4 and S4). The value discrimination activities of neurons might influence subsequent behavior, resulting in an observable correlation between neural activity and the second finger insertion (Fig. 2c, d). To examine how the neural activity changed as finger insertion behavior adapted to value reversals, we analyzed the correlation between RTs for the second finger insertion and the activities of value neurons during stimulus presentation and the delay period (Figs. 5a, b and S9).

When the value of each stimulus shifted from good to bad, the speed of finger insertion gradually decreased, possibly as the monkeys recognized the change in stimulus value (from 370.18 ± 1.77 ms for good stimuli to 996.22 ± 8.87 ms for bad stimuli, based on averaged data from both the T-VRT and V-VRT tasks, mean ± SE) (Fig. 5a, b). This alteration in behavior was accompanied by a dynamic increase in the neural responses of negative value-coding neurons (Fig. 5a, b, left panels). In contrast, the neuronal responses of positive value-coding neurons decreased as the monkeys began to insert their fingers more slowly after the value shifted from good to bad (Fig. 5a, b, right panels). Similar correlations were observed when the value shifted from bad to good (Fig. S9). These results show that the change in finger insertion speed following the value switch coincided with changes in the neural responses of value neurons, suggesting their potential role in guiding finger insertion driven by anticipated value.

We found two types of value neurons classified by when they encoded values: stimulus and delay value neurons (Fig. 4). Considering that the responses of the delay value neurons aligned with the second finger insertion more closely than the other type, it raises the question of whether they play distinct roles in guiding anticipated value-driven finger insertion behavior. To determine this, we calculated the

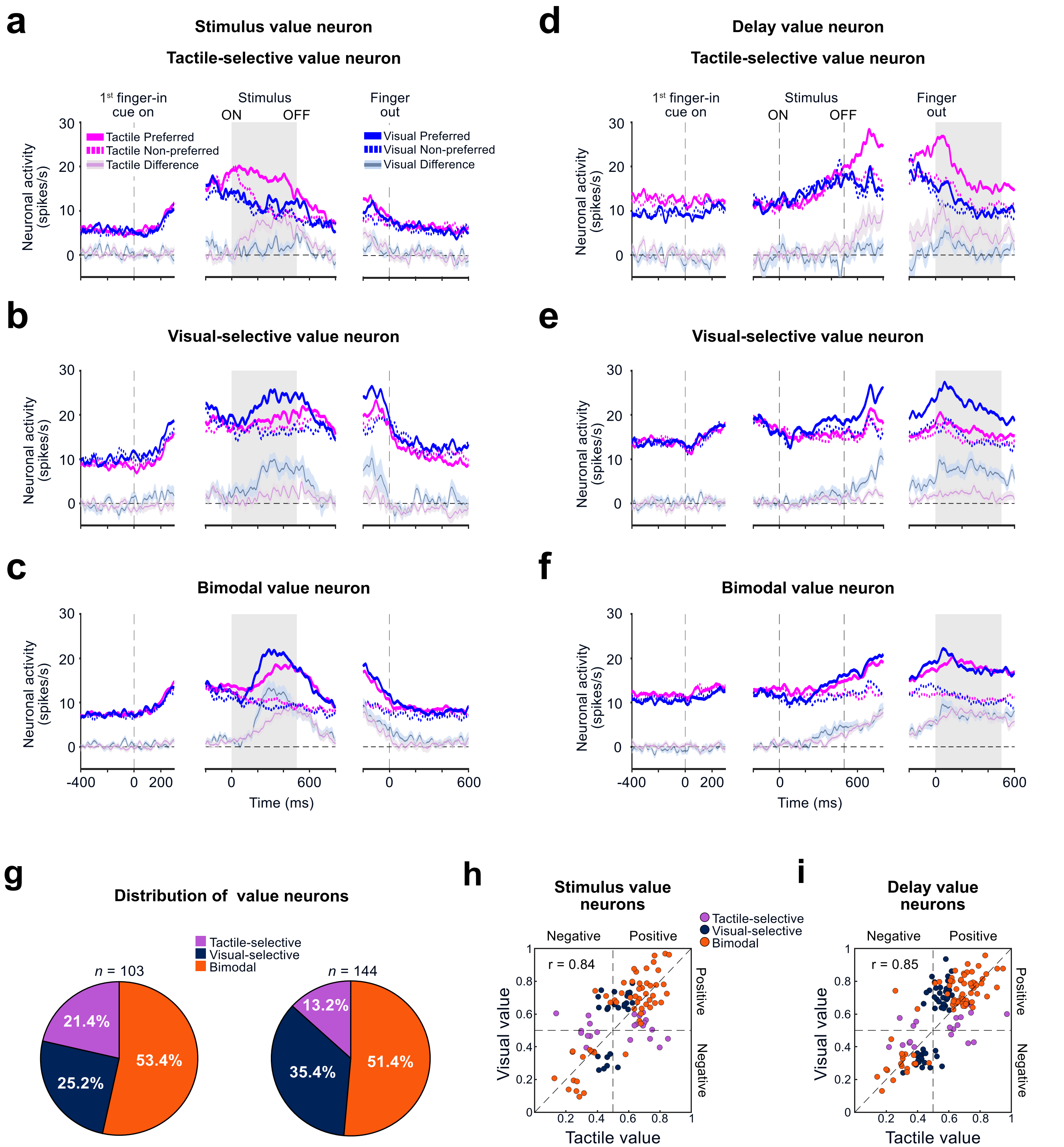

Fig. 4 | Half of neurons encoded both values, and the other half encoded modality-selective values. **a–c** The population responses of each type of value neuron in the putamen during the stimulus presentation period. The peristimulus time histograms (PSTHs) indicate the averaged activities of tactile-selective ($n = 22$) (**a**), visual-selective ($n = 26$) (**b**), and bimodal ($n = 55$) (**c**) value neurons to good and bad stimuli during the T-VRT and V-VRT. The value discrimination activities in each task are indicated by transparent lines (mean ± SE). **d–f** Population responses of each type of value neuron in the putamen during delay period. The same format as in (**a–c**) for tactile-selective ($n = 19$) (**d**), visual-selective ($n = 51$) (**e**), and bimodal ($n = 74$) (**f**) value neurons. **g** Distribution of the three types of value neurons in the putamen. The left and right pie charts show the distributions of stimulus and delay value neurons, respectively. **h** Comparison between the tactile value coding (abscissa) and visual value coding (ordinate) of stimulus neurons in the putamen ($n = 103$). For each neuron (each dot), the magnitudes of the tactile and visual value coding, calculated by ROC areas, are plotted. ROC areas higher and lower than 0.5 indicate positive and negative value coding, respectively. **i** Comparison between tactile value coding and visual value coding of delay neurons in the putamen ($n = 144$). The format is the same as in (**h**).

correlation between the neural activities of each type of value neuron and individual finger-in RTs. Figure 5c shows an example result: 75.68% of bimodal delay value neurons exhibited statistically significant correlations between their responses during the delay period and the RTs for the second finger insertion ($p < 0.05$, Pearson correlation coefficient). In contrast, only 12.16% of these same neurons showed correlations between their responses during the stimulus period and the RTs.

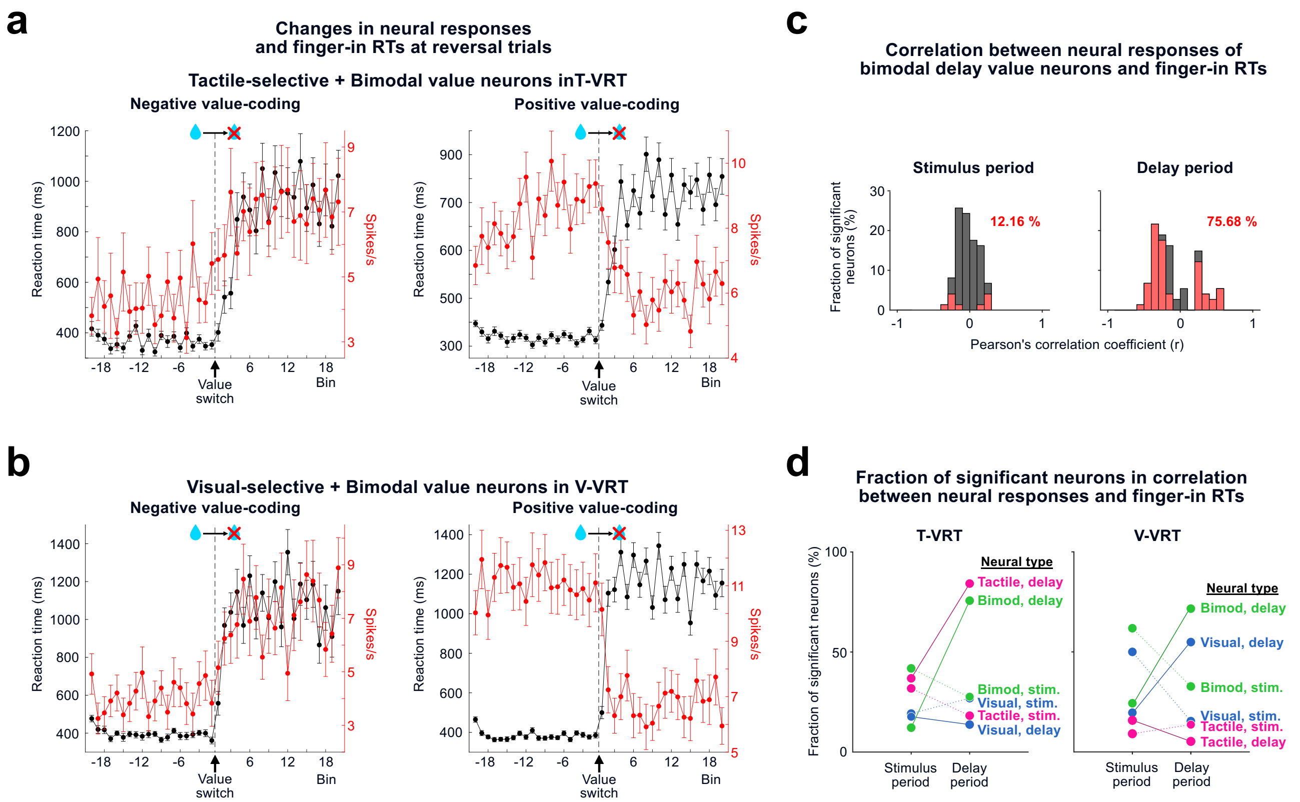

Fig. 5 | Neural responses correlated with anticipated value-guided finger insertion. a Changes in neural responses and finger insertion speeds across trials in the tactile value reversal task (T-VRT). The averaged spike rates of tactile-selective and bimodal value neurons (red) and reaction times of finger insertion in response to the second finger-in cue (black) are plotted before and after the value switching trials (mean ± SE, with a maximum and minimum sample size for each data point: 60 and 40 for negative value-coding, and 126 and 85 for positive value-coding). **b** Plot with the same format as in (**a**) for visual-selective and bimodal value neurons in the visual value reversal task (V-VRT) (mean ± SE, with a maximum and minimum sample size for each data point: 60 and 40 for negative value-coding and 63 and 36 for positive value-coding). **c** Example correlation coefficient histograms of bimodal delay value neurons in the T-VRT ($n = 74$). The correlation coefficient values were calculated by comparing between the reaction times of finger insertion and neural responses during the stimulus presentation period (left) and delay period (right). Red bars indicate neurons with statistically significant correlations (two-tailed Pearson correlation coefficient, threshold for significance: $p < 0.05$). The percentage indicates the fraction of neurons showing statistically significance in correlation. **d** Dynamic changes in the fractions of value neurons correlated with anticipated value-guided finger insertion. The fractions of 6 types of value neurons showing significant correlation coefficients during the stimulus and delay periods are plotted separately for the T-VRT (right) and V-VRT (left).

All types of value neurons were analyzed to calculate their correlations with finger-in RTs (Fig. S10). In the T-VRT, the proportions of tactile and bimodal delay value neurons showing statistically significant correlations between their neural responses and finger-in RTs selectively increased during the delay period, compared with the stimulus period (from 36.84% to 84.21% for tactile selective; from 12.16% to 75.68% for bimodal) (Fig. 5d, left panel). Similarly, in the V-VRT, the proportions of visual and bimodal delay value neurons selectively increased during the delay period compared with the stimulus period (from 19.61% to 54.9% for visual selective; from 24.32% to 71.62% for bimodal) (Fig. 5d, right panel). These findings suggest that delay value neurons play a predominant role in guiding the finger insertion behavior driven by anticipated values.

Efficient processing of value information by bimodal value neurons

Our finding of convergent representation of values from tactile and visual inputs leads to a fundamental question: What advantages do the bimodal value neurons offer for encoding values? One potential advantage is requiring fewer neurons to encode values, compared with parallel processing. Because bimodal value neurons can participate in generating both tactile and visual value discrimination responses, we directly simulated that the total number of value neurons decreases as the number of bimodal value neurons increases to discriminate value at the single-neuron level (Fig. 6a). In this value encoding process with bimodal value neurons, the number of neurons dedicated to processing each value remains consistent with that in parallel processing, even when using a smaller total number of neurons.

In addition, to test whether the presence of bimodal value neuron also facilitates the efficiency of value processing at the population level of value discrimination, we conducted simulations using a population decoding analysis across various configurations of value neurons[31]. The value-decoding accuracies were first compared between two models, each comprising 100 neurons: the parallel processing model (1:0:1 for tactile-selective, bimodal, and visual-selective neurons) and the convergent processing model (1:2:1) (Fig. 6b). Notably, value classification accuracies were higher in the convergent processing model than the parallel processing model in both tasks (Fig. 6b). Next, we calculated the maximum accuracies as the count of value neurons increased in these two models (Fig. 6c). Our simulations consistently revealed that the 1:2:1 configuration of convergent processing outperformed parallel processing, achieving a higher accuracy with a lower number of value neurons (Fig. 6c).

To investigate the efficiency of parallel processing in encoding values as the number of bimodal value neurons increases, we calculated the number of neurons needed to achieve an 80% value-decoding accuracy across different configurations of value neurons. Those

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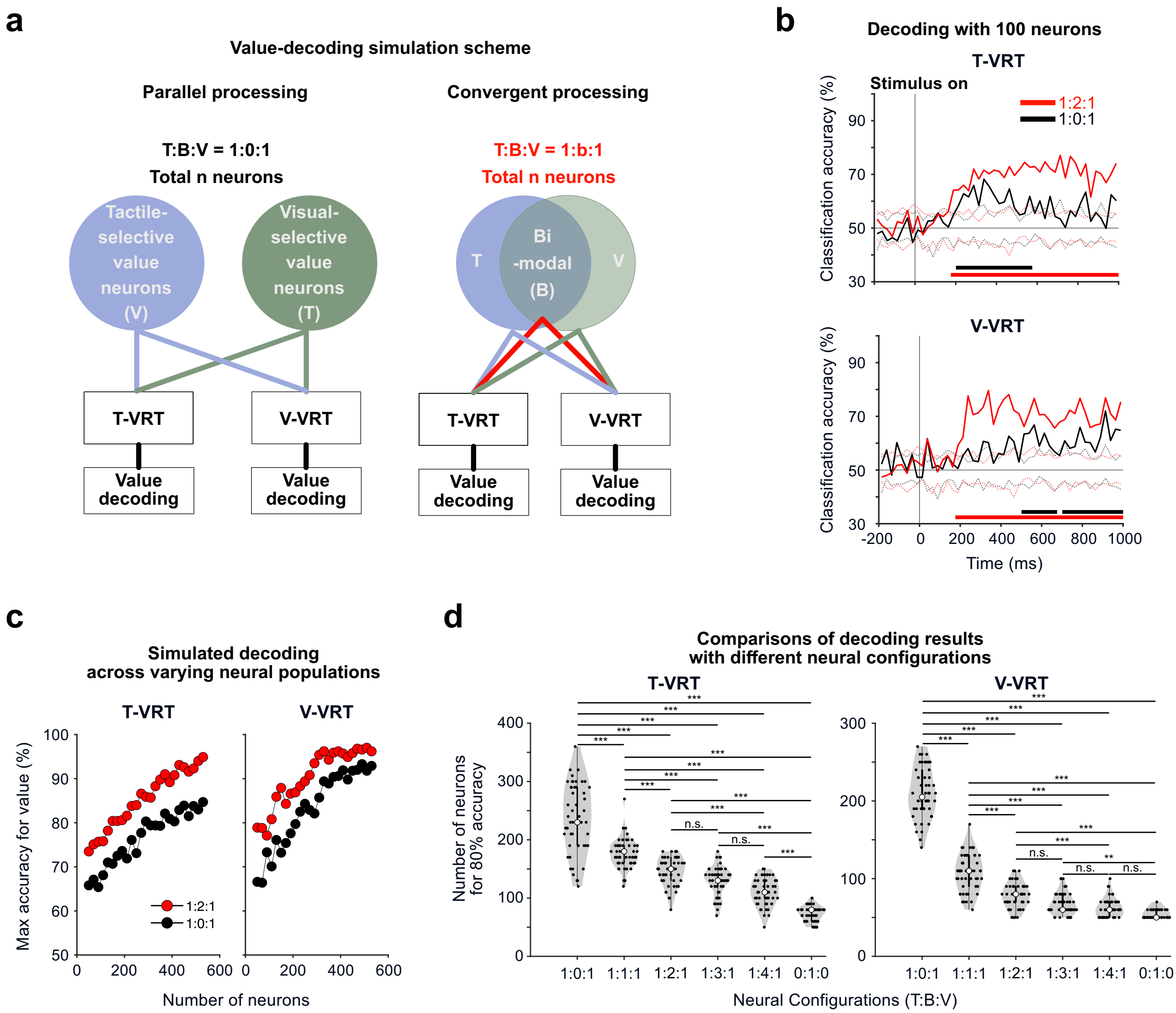

Fig. 6 | Simulation of parallel and convergent value processes using modality-selective and bimodal value neurons. a Simulation scheme for value decoding with various neural configurations representing the parallel and convergent processes. Value decoding accuracies were calculated with neural population activities with bimodal value neurons (convergent process, right) and without them (parallel process, left). **b** Examples of value decoding in parallel and convergent processes. Value decoding accuracies in the parallel value process (1:0:1 ratio of tactile, bimodal, visual value neurons; $n = 100$) and convergent value process (1:2:1; $n = 100$) were compared in the tactile value reversal task (T–VRT) and visual value reversal task (V-VRT). The black and red dashed lines indicate the decoding accuracies of the permutation test. Bars represent periods of decoding accuracy above the permutation results. **c** Example plots showing the maximum values of decoding accuracy calculated with various total numbers of value neurons in the T-VRT (left) and V-VRT (right). **d** Comparisons of the minimum number of neurons needed to achieve an 80% decoding accuracy among 6 possible neural configurations in the T-VRT (left) and V-VRT (right). The solid thick black bar within each violin plot represents the inter-quantile range (from the 1st to the 3rd quantile) of the data points. The white circle within each plot indicates the median value. One-way ANOVA was applied for comparison ($n = 50$, $F(5,294) = 156.77$, $p = 9.735 \times 10^{-81}$ for T-VRT and $F(5,294) = 466.47$, $p = 1.811 \times 10^{-137}$ for V-VRT; post hoc Bonferroni pairwise comparison; exact p values for each comparison are provided in the source data file). $^{*}p < 0.05$, $^{**}p < 0.005$, $^{***}p < 0.0005$.

simulation results revealed that an increase in the proportion of bimodal value neurons significantly reduced the total number of value neurons required to achieve an 80% value-decoding accuracy (one-way ANOVA and post hoc Bonferroni pairwise comparison, $p < 0.05$) (Figs. 6d and S11). Our data suggest that as value convergence increases, the efficiency of value encoding improves. However, we did not find significant differences in modality-decoding accuracies between convergent and parallel processing (one-way ANOVA) (Fig. S12).

Overall, our simulation results at both the single-neuron and population levels demonstrate that the use of convergent neural processing with bimodal value neurons enables the putamen to encode value information with fewer neurons than would be required for parallel processing.

Discussion

Our fMRI study demonstrates that object values are represented in the human putamen, independently of tactile or visual input. Interestingly, our single-unit recording in monkeys reveals that more than half of individual putamen neurons encode both tactile and visual values, while the other half selectively encodes either value. These findings indicate that values originating from different sensory modalities are processed within single neurons in the putamen. The presence of these bimodal value neurons in the putamen further suggests that the anatomical funneling structure of the basal ganglia is not merely structural but also functional, efficiently extracting values from both tactile and visual inputs through a convergent process. We further address the question of whether bimodal value neurons provide any advantage in value processing. Our simulation results show that this convergent value process with the bimodal value neurons uses fewer neurons for value discrimination than the parallel value process at both the single-neuron and population levels. Thus, our data indicate that the primate putamen plays a crucial role in convergent processing of values from tactile and visual inputs, enhancing the efficiency in value-guided finger insertion behavior.

Tactile value processing in putamen neurons for decision-making via the fingertip

Our study discovers that the putamen neurons encode the values associated with tactile stimuli experienced at the fingertip. Traditionally recognized for its role in motor control and action value processing, the putamen's role in processing object value has received relatively little attention[32–34]. In addition, a previous macaque study showed that the caudal region of the putamen encoded stably sustained value of visual object for habitual behavior, but the process of flexibly updating value in the primate putamen remained unclear[18]. Notably, our study reveals that neurons in the putamen flexibly encode changing values of visual and tactile stimuli, suggesting its role in flexible value processing. These findings highlight a previously uncharted role for the putamen in encoding flexible values from tactile and visual inputs through reward prediction error (RPE) signals from dopamine neurons, enabling primates to make optimal decisions by collecting information via their fingertips and eyes.

In our fMRI results, while value discrimination in the putamen was observed during both the tactile and visual tasks, value discrimination in the ventral striatum (VS) was detected only during the visual task. However, we need to be cautious not to over-interpret these null results, as they do not necessarily indicate that tactile value information is not processed in the VS, based on prior research supporting the processing of reward-related information across various sensory stimuli[35,36]. If most good value-coding neurons are localized very close to bad value-coding neurons in the VS, detecting value discrimination through multivoxel pattern analysis of fMRI data would be challenging. As it is known that multiple brain structures, including the caudate and ventral striatum, are involved in the decision-making process based on value, studying the putamen alongside these regions is crucial for understanding convergent and parallel value processes at the whole-brain level[17,37]. Therefore, examining neural responses in the three striatal regions, the putamen, caudate, and ventral striatum, and their input and output structures will enhance our understanding of efficient decision-making mechanisms at the whole-brain level in the future.

Integration of tactile and visual values in putamen neurons

How do individual neurons in the putamen process both tactile and visual value information? Various funneling models have previously been suggested and debated, and our findings provide insight into how neurons converge these distinct values[7,10]. Neurons in the putamen receive diverse sensory inputs, including tactile and visual inputs, directly from cortical areas[21,38–40]. Moreover, our functional connectivity results from the human fMRI data indicate that the putamen interacts with sensorimotor areas in a modality-dependent manner (Fig. S13). Furthermore, dopamine inputs from the substantia nigra pars compacta strongly innervate the putamen, enabling it to encode values from various sensory modalities[41].

These anatomical studies suggest that a potential circuit of axon terminals from individual neurons in the cortex that are responsible for processing either tactile or visual information establishes connections with single neurons in the putamen. Dopamine projection to the putamen might modulate the synaptic strength of these convergent inputs during value learning. This neural pathway allows for the integration of value information from both sensory modalities in single neurons of the putamen. In contrast, putamen neurons encoding either tactile or visual value might receive a single input from the relevant modality-processing area in the cortex. The same latencies for value discrimination responses of bimodal and modality-selective value neurons suggest that the value may be influenced by dopamine inputs with similar latencies of RPE signals (Fig. S8). However, sensory responses to the stimuli between bimodal and modality-selective value neurons appear to be different in their response patterns, suggesting that tactile and visual sensory inputs into the putamen may be processed differently (Fig. S14).

In addition, these anatomical circuits and heterogenous functions of dopamine neurons raise two possibilities for how dopamine neurons update value: dopamine neurons may update the value information regardless of the modality system, or dopamine neurons may be grouped to update each modality value[30,42]. Interestingly, it is known that dopamine neurons receive different kinds of sensory inputs, including vision, touch, and sound, from the pedunculopontine tegmental nucleus (PPTg)[43]. Moreover, we found that the RPE signal was reflected by neuronal responses in the putamen. Bimodal value neurons exhibited a statistically significant increase in neural response to unexpected rewards and a decrease in response to unexpected non-rewards, compared to baseline activities before the outcome deliveries (two-tailed Wilcoxon rank-sum test, $p < 0.05$) (Fig. S15). However, modality-selective value neurons did not exhibit statistically significant changes. Our data suggest the possibility of different value processing mechanisms between bimodal and modality-selective value neurons. Since no study has shown whether these sensory inputs converge on a single dopamine neuron or not, this is an important future study to reveal how primate dopamine neurons update value from various sensory inputs.

Overall, our results indicate that both parallel processing and convergent processing occur in the putamen. This raises the important future question of whether the successive structures, such as the globus pallidus and substantia nigra, and the striatum, including the ventral striatum and caudate, handle these values through the convergent funneling process. Studying how tactile and visual value information is processed in the upstream and downstream structures of the putamen will enhance our understanding of the overall information flow in the brain.

Advantages and disadvantages of convergent value processing through bimodal value neurons

An important question is how the brain efficiently processes vast amounts of information with a limited number of neurons, a crucial aspect of natural intelligence in contrast to artificial intelligence, which can easily increase its computing resources. Our findings provide empirical evidence that convergent processing with bimodal value neurons requires fewer neurons to represent values than parallel processing. This, in turn, might conserve resources and increase the brain's capacity for processing other types of information.

On the other hand, this convergent process seems to retain only value information, omitting sensory modality-specific details, implying that each bimodal value neuron encodes unified value. Consequently, it becomes challenging to discriminate which modality is processed based on the averaged responses of bimodal value neurons (Fig. 4c, f). However, we observed that the neural response patterns preserved this modality information, as shown in the decoding analysis (Fig. S12). This finding suggests that putamen neurons might use their population response patterns to preserve the compressed representation of sensory modalities[8].

Understanding which neuronal properties contribute to these processing differences is also an important issue. Activity variabilities between bimodal value neurons and other types of value neurons showed no significant difference, meaning that the difference in value processing efficiency is not directly driven by the consistency with which value information is encoded (Fig. S3e). Additionally, a neuron group composed solely of bimodal value neurons required less than half the number of neurons to achieve the 80% decoding accuracy compared to a group with only modality-selective value neurons (Fig. 6d). This suggests that the efficient decoding facilitated by adding bimodal value neurons is not simply due to the number of neurons involved in value processing; rather, the quality differences of neural population responses may be involved.

An intriguing observation is that the decoding weights of each neuron were highly correlated with their value ROCs, indicating that neurons with clearer value discrimination responses (greater distance of value ROC from 0.5) more strongly contributed to the decision boundary of the decoder (greater distance of neural weight from 0) (Fig. S16). Importantly, a substantial number of bimodal value neurons had large absolute neural weights, suggesting their stronger role in shaping the decision boundary of the linear decoder compared to modality-selective value neurons. Our analyses suggest that the enhanced decoding performance observed with an increasing number of bimodal value neurons may be due to their greater contribution to the decoding process. However, comprehending how single-cell properties impact population representation is a complex issue that remains a challenge in neuroscience[44–47]. It is also plausible that various other properties of bimodal value neurons could have contributed to the enhanced population performance. Many questions remain in neuroscience regarding how neural population patterns function, process information, and make decisions[48–51].

Possible roles of the parallel value process through modality-selective value neurons

Our study also shows that nearly half of putamen neurons selectively process either tactile or visual value, suggesting the presence of parallel processing through the putamen. These modality-selective value neurons could serve three possible functions.

First, modality-selective value neurons could act as markers of sensory modality. In cases in which bimodal value neurons exclusively process both tactile and visual values, it becomes challenging to differentiate the sensory source based solely on population average responses. Each modality-selective value neuron might function as a distinct identifier for its associated sensory modality, simplifying the interpretation of sensory information. Second, these neurons might form selective connections with specific motor output regions. For example, visual- and tactile-selective value neurons in the putamen could project to distinct areas controlling eye and forelimb movements, respectively, in the globus pallidus internal segment[6,52]. These modality-selective circuits could be involved in guiding a particular movement, such as saccades or grabbing. Third, these modality-selective value neurons come into play when conflicting value information arises from two sensory modalities. In such value conflicts, bimodal value neurons might struggle to differentiate between conflicting values from different modalities. Conversely, each type of modality-selective value neuron might independently process its specific value to guide behavior when value conflicts occur.

Transitioning from value recognition to finger-in action: the possible roles of stimulus and delay value neurons in prefrontal-striatal circuits

In our value tasks, we temporally separated the phases for value recognition and finger-in preparation. This allowed us to identify two populations of neurons in the putamen: stimulus and delay value neurons. Notably, we found a small number of neurons that sustained their value discrimination responses from the stimulus presentation period to the delay period (10.12%, 25 out of 247 value-coding neurons). Most value-coding neurons exhibited a stronger preference for encoding values during either the stimulus presentation or delay period, indicating their predominant involvement in one of these periods (Fig. 4a–f).

Interestingly, previous studies have reported the presence of two types of neurons in the prefrontal cortex (PFC) that showed stronger responses during either cue presentation or the delay period[53–56]. Considering these two types of PFC neurons and the cortico-striatal connection, it is possible that the distinct types of value neurons in the putamen might connect to the prefrontal regions through different pathways, potentially serving distinct functions. Moreover, our additional functional connectivity results from human fMRI data, using the putamen as the seed region, showed different strengths of connections with the inferior frontal cortex and the lateral orbitofrontal cortex depending on modality and value conditions (Fig. S17). This suggests that distinct prefrontal circuits and computational mechanisms may be involved in processing different modalities and values. Future investigations, particularly those involving simultaneous recordings of the prefrontal regions and putamen, will be crucial to elucidate the circuit-level mechanisms underlying value-guided actions through the convergent and parallel processing of tactile and visual information.

Methods

fMRI study with human participants

Participants. Twenty-two neurologically intact right-handed participants (mean age 24.36 ± 3.58 years, range 20–33, 10 females) took part in the experiment. The participants reported that they had normal or corrected-to-normal vision. All participants provided informed consent for the procedure. The experimental procedure was approved by the Institutional Review Boards of Seoul National University and the Korea Advanced Institute of Science and Technology.

Stimuli. As tactile stimuli, we used two braille patterns engraved on the surfaces of polyoxymethylene blocks (Fig. 1a). The dimensions of each tactile block were 18 × 18 × 12 mm, and each dot had a diameter and height of 2 × 1 mm. The tactile stimuli were automatically presented via a pneumatic tactile stimulus delivery system, as previously reported[57]. For the visual stimuli, we used two fractal images created with Fractal Geometry (Fig. 1a)[25,58]. The mean luminance across the images was equalized using the Spectrum, Histogram, and Intensity Normalization and Equalization (SHINE) toolbox written with MATLAB (www.mapageweb.umontreal.ca/gosselif/shine). Grayscale images of the fractal objects were used to minimize the color effect on brain responses. The fractal images were presented with a visual angle size of 7 × 7 degrees.

Task design: value reversal tasks for human participants. Participants performed two distinct value reversal tasks, the tactile task and the visual task, while their neural activities were monitored using fMRI. Each task was executed in separate runs, with participants being informed about the upcoming task before each run. In both tasks, tactile and visual stimuli were presented concurrently. Specifically, the tactile task exclusively linked tactile stimuli, the braille T-a and T-b, with monetary values, with one stimulus predicting monetary gain (good), and the other predicting monetary loss (bad) (Fig. 1b). The visual stimuli in the tactile task were unrelated to the monetary outcome. Conversely, in the visual task, the fractal images were associated with monetary values. One image was linked to monetary gain (good), and the other was associated with monetary loss (bad), and the braille stimuli were not linked to the outcomes (Fig. 1b). The rule linking each stimulus to its corresponding monetary value was switched in the middle of each run in both tasks. Each task comprised four scan runs, with trial counts of 16, 18, 18, or 20 for each run. The order of the runs was pseudo-randomized for each participant. The variation in trial numbers within each run was intended to mitigate participants' ability to predict the occurrence of rule switching. The tactile task and visual task alternated, and the task modality of the first run was counterbalanced across participants.

In each trial of both tasks, the onset of the trial was indicated by the presentation of a blue fixation cross. After 500 ms, the cross changed to red as the "insert finger" signal (Fig. 1c). Participants were instructed to insert their right index finger into the entrance hole and touch a braille stimulus in the hole when the red cross was displayed and to withdraw their finger from the hole when the red color of the fixation cross returned to blue ("withdraw finger" signal) 1 s later (Fig. 1c). Participants were concurrently exposed to visual stimuli while experiencing the tactile stimuli. To ensure equal exposure to both tactile and visual stimuli, the visual stimulus was presented only while the finger was in the entrance hole. After a 5 s interval, participants were asked to decide whether to take the object, contingent upon its potential value derived from feedback in preceding trials or through conjecture. To mitigate the potential interference of pre-planned motor signals with value representation, the plus (indicating "take") and minus (indicating "do not take") buttons were randomly assigned to the upper and lower positions (Fig. 1c). Participants were instructed to promptly press the upper button with their left index finger and the lower button with their left middle finger. The response window lasted for 2 s, followed by a 0.5 s feedback period and a variable inter-trial-interval ranging from 2.5 to 13 s.

As feedback, the monetary gains or losses associated with each trial's response were displayed just above the fixation cross (Fig. 1c). Before the tasks, the participants were informed that the presented feedback would be added to their compensation after the experiment. When the presented stimulus was associated with monetary gain, selecting the plus button resulted in a gain of 150 won, and selecting the minus button yielded a gain of 50 won. If the presented stimulus was linked to monetary loss, choosing the minus button led to a loss of 50 won, and selecting the plus button resulted in a loss of 150 won. Participants were familiarized with these responses and feedback rules before the task.

Before undertaking the two value reversal tasks, participants underwent a preparatory session inside the scanner to familiarize themselves with the stimuli and the finger movement protocol. This familiarization session encompassed four tactile blocks and four visual blocks with four braille patterns and four fractal images, inclusive of the stimuli used in the subsequent value reversal tasks. The participants performed alternating tactile and visual blocks. During the tactile block, the participants were instructed to insert their right index finger into the entrance hole to touch and discern the given tactile stimulus while they fixated on the center of the screen. During the visual block, the participants were asked to view each fractal image displayed at the center of the screen while tactually exploring a plain surface with their right index finger. Throughout the familiarization session, each stimulus was presented ten times, including one-back response trials (20% of the total trials), wherein participants were required to determine whether the tactile (or visual) stimulus of the current trial matched that of the previous trial. Additional details on this familiarization protocol are available in our previous work[57].

fMRI acquisition. Participants underwent scanning on a 3T Siemens MAGNETOM Trio located at Seoul National University. Echo-planar imaging (EPI) data were acquired using a 32-channel head coil with an in-plane resolution of 2.8 mm × 2.8 mm and 40 2.5 mm slices (0.25 mm inter-slice gap). The scanning parameters were a repetition time (TR) of 2000 ms, echo time (TE) of 25 ms, matrix size of 76 × 76, and field of view (FOV) of 210 mm. Whole brain volumes were scanned, and slices were oriented approximately parallel to the base of the temporal lobe. Following the experimental runs of the fMRI session, anatomical images were acquired using the standard magnetization-prepared rapid-acquisition gradient echo (MPRAGE) sequence.

fMRI analysis. fMRI data were analyzed using AFNI (http://afni.nimh.nih.gov), SUMA (AFNI surface mapper), FreeSurfer, and custom MATLAB scripts. Standard data preprocessing procedures, including slice-time correction and motion correction, were conducted.

To identify regions encoding value information within the human striatum, event-related brain activation patterns in the BOLD signal were initially extracted at the onset of tactile and visual stimulus perception in each trial. This extraction was performed using a standard general linear model in the AFNI software package (3dDeconvolve using the GAM option with motion parameters), and individual trials were modeled as separate events. The estimated t-value for each trial was used in the searchlight analysis[59]. Searchlight spheres, each with a radius of 9.8 mm (corresponding to approximately 123 voxels) were defined within the striatum mask. This mask encompassed the bilateral caudate, putamen, and nucleus accumbens, automatically delineated by the parcellation of FreeSurfer ("aparc.a2009s+aseg_REN_all.nii.gz"). The t-values extracted from the voxels within each sphere across the striatum were then normalized within each voxel by subtracting the mean value across all stimulus conditions. To derive value discrimination indices, Pearson correlation coefficients were calculated between the voxel patterns of within-value trials (good-and-good or bad-and-bad) and those of between-value trials (good-and-bad). These correlation coefficients underwent Fisher's z transformation, and the value discrimination indices were defined as the mean within-value correlations subtracted by the mean between-value correlations. When calculating discrimination indices, trials involving incorrect button responses, the first trial of each run, and rule-switching trials were excluded, and all remaining possible pairs of trials from different runs were used. The results were registered to the MNI space from each participant's original volume for the subsequent group analysis.

Statistical analysis. To compare behavioral performance between the tactile and visual tasks, a paired t-test (two-tailed) was conducted in MATLAB. To assess the significance of the value discrimination indices against the baseline level (zero), one-sample t-testing (right-tailed) was used through the AFNI software function (3dttest++).

Electrophysiology study with macaque monkeys

Braille presentation for non-human primates. We developed an apparatus for presenting braille patterns to non-human primates (Figs. 2a and S1). The braille presenter comprises pairs of four main components: pneumatic pin cylinders, braille units, photo interrupters as finger sensors, and endoscopic cameras (Fig. S1a, b).

Each braille presentation case includes a finger entrance hole to allow tactile exploration of the braille pattern presented inside (Fig. S1b). A braille unit consists of six dots, each independently controlled by an individual pin cylinder placed below the braille unit (Fig. S1a). Each pin cylinder vertically moves a wire connected to a braille dot through air pressure delivered by a compressor (KC-T002, Korea). The computer program BLIP, designed for electrophysiology and behavior studies with macaque monkeys (National Institutes of Health, USA), was customized to control the movement of each braille pin in accordance with the task schedule[25,57].

To precisely measure the timing of finger insertion into the hole in the braille presentation case, two photo interrupters (F249, Arduino module) were placed on opposite sides of the finger entrance hole (Fig. S1b). To further ensure real-time monitoring of the monkeys' finger insertion movements and braille pattern presentation, endoscopic cameras (PS-EC200, Korea) were installed at each finger entrance hole (Fig. S1c). We additionally monitored the monkeys' arm movements during the experiment using a camera (D455, Intel, USA) installed above the braille presenter.

Visual stimuli. We used two fractal images created using Fractal Geometry for the V-VRT[25,58] (Fig. 2b). The mean luminance was equalized across the images using the SHINE toolbox written with MATLAB (www.mapageweb.umontreal.ca/gosselif/shine).

Tactile stimuli. Braille patterns were used as the tactile stimuli in the T-VRT. Each braille unit consists of six dots, which were made of an aluminum alloy (Fig. S1b). The diameter and height of each dot was 2 × 1.5 mm, and the size of each braille unit was 10 × 5 mm.

General procedures. Two adult monkeys (*Macaca mulatta;* 5.2 kg female monkey EV and 10.7 kg male monkey UL) were used for the non-human primate experiments. Animal care and experimental procedures were approved by the Seoul National University Institutional Animal Care and Use Committee. Under general anesthesia and in surgical conditions, a plastic head holder and a recording chamber were implanted onto the monkeys' skulls. The position of each chamber was tilted laterally by 25° so that it corresponded to the putamen. We started the training and recording sessions after the monkeys were fully recovered from the surgery.

Behavioral tasks. The behavioral tasks were controlled by a Windows-based experimentation data acquisition system, BLIP (National Institutes of Health, USA) (www.cocila.net/blip). The monkey sat in a primate chair facing a fronto-parallel screen in a sound-attenuated and electrically shielded room. The visual stimuli, squares and fractals, were approximately 20° × 20° in size. These stimuli were presented on an LCD monitor with a 120-Hz refresh rate (29WL500, LG, Korea). An infrared camera was used to record the monkey's eye position with a sampling rate of 1 kHz (Bio-signals, USA). The BLIP software precisely monitored the timing of visual stimulus presentation using a signal from a photodiode attached to the monitor. Tactile stimuli in the form of braille patterns were presented using the braille presenter, as described above.

Tactile value reversal task and visual value reversal task for macaque monkeys. To test whether individual neurons converged values from different sensory modalities, we designed tasks in which each tactile or visual stimulus could be associated with a liquid reward outcome (Figs. 2c, d and S2b, c). The two tasks using visual (V-VRT) and tactile (T-VRT) modalities were introduced while recording a single neuron. Each task was composed of single-stimulus trials and choice trials.

To examine tactile and visual value responses in individual neurons, we used a value reversal procedure in which the stimulus-value contingency was reversed in each block of 50–60 trials (Fig. 2b). In the first block of T-VRT or V-VRT, one stimulus was associated with a liquid reward, and the other was not. In the subsequent block, the object-reward contingency was reversed. The selection of the stimulus associated with a liquid reward in the initial block was randomized.

T-VRT procedure. In the single-stimulus T-VRT trials, the monkeys were trained to insert their left index finger into the left hole of the braille presentation case to experience the braille pattern when the finger-in cue appeared on the screen (colored square cues). To ensure precise exposure to the tactile stimulus during the first stimulus presentation, one of two braille patterns (1-dot and 6-dot) was delivered to the monkeys 50 ms after their index fingers entered the hole, and the delivered braille pattern was maintained for 500 ms (Fig. S2a). The monkeys were required to maintain contact with the tactile stimulus until the finger-in cue disappeared along with the stimulus. The monkeys had to rely solely on recognizing the value of the tactile stimulus with their tactile perception because the braille pattern presented in the opaque braille presentation case was not visible. After finger withdrawal, a blank black screen was presented for a randomized duration ranging from 500 to 1000 ms.

When the second finger-in cue appeared on the screen, the monkeys could insert their finger back into the hole to touch the same tactile stimulus experienced during the first finger insertion. However, the monkeys were free to decide whether to insert their finger or not when the second finger-in cue was presented. To receive the reward when the stimulus associated with a liquid reward was presented, the monkeys were required to maintain their finger insertion for 200 ms. The reaction time for finger insertion was measured as the time from the presentation of the second "finger-in" cue to the actual finger insertion (Fig. S2a).

A choice trial followed every four single-stimulus trials. In the T-VRT choice trial, two square cues for choice were presented on the screen, and two different braille patterns were simultaneously displayed in the two holes in the braille presentation case. The monkeys were trained to freely explore these two braille patterns by inserting their left fingers into each hole. To choose one stimulus, monkeys had to maintain their finger insertion for more than 1500 ms. The reward was provided after the monkeys completed the choice finger insertion (Figs. 2c, d and S2b, c, lower scheme). For monkey UL, the choice square cues were simply presented at the beginning of the trials (Fig. 2c, d). For monkey EV, however, the choice trial commenced with the first "finger-in" cue instead of two choice cues. EV was required to insert a finger into the hole for 500 ms. After finger withdrawal and the delay period, two square cues for choice were presented on the monitor (Fig. S2b, c).

V-VRT procedure. The procedure for the V-VRT was the same as for the T-VRT, except for the presentation of visual stimuli instead of tactile stimuli (Figs. 2c, d and S2b, c). To ensure consistent finger movements in both the T-VRT and V-VRT, the monkeys were also trained to insert their left index finger to display the fractal images in the V-VRT. Additionally, to minimize experimental confounding factors from the braille pattern-raising acoustic noise, a dummy braille pattern in the unused right hole was presented during the single-stimulus trials. This ensured that monkeys experienced a similar level of noise in both the V-VRT and T-VRT.

In the single-stimulus trials, the first finger-in cue was replaced with one of two fractal images by finger insertion. The monkeys were required to keep inserting their finger for 500 ms. After finger withdrawal and the delay period, the second "finger-in" cue appeared on the screen. The monkeys were allowed to freely decide whether to insert their fingers or not upon the second cue. When the monkeys reinserted their finger into the hole, the same fractal image presented during the first finger-in period was displayed for 200 ms. The liquid reward was delivered 200 ms after finger reinsertion.

In the choice trials, two square cues for choice were presented on the screen. The monkeys were free to insert their fingers into the holes to choose one fractal image. Each fractal image was presented to the monkey after they inserted their finger into the hole. The monkeys selected a fractal image by maintaining finger insertion for 1500 ms. The reward was delivered upon completion of the monkeys' choices. The overall procedures for the choice trials for each monkey were the same as those in the T-VRT.

Single-unit recording. While the monkey performed a task, the activity of single neurons in the putamen was recorded using conventional methods. The recording sites were determined using a 1 mm spacing grid system, with the aid of MR images (3T, Siemens) obtained along the direction of the chamber. Single-unit recording was conducted using a glass-coated electrode (Alpha-Omega). The electrode was inserted into the brain through a stainless-steel guide tube and advanced by an oil-driven micromanipulator (MO-974A, Narishige). Neuronal signals from the electrode were amplified, filtered (250 Hz to 10 kHz), and digitized (30-kHz sampling rate and 16-bit A/D resolution) by a Scout system (Ripple Neuro, UT). Neuronal spikes were isolated online using custom voltage-time window-discrimination software (BLIP, Laboratory of Sensorimotor Research, National Eye Institute

National Institutes of Health, available at www.cocila.net/blip), with the corresponding timings detected at 1 kHz. The waveforms of individual spikes were collected at 50 kHz.

Data analysis

Behavioral analysis. To assess the speed of finger insertion into the hole based on anticipated value, we calculated the RT from the presentation of the second "finger-in" cue to the actual finger insertion. The statistical significance of RT differences between good and bad stimuli was validated with a two-tailed unpaired *t*-test. To illustrate the RT differences across all neural recording sessions, they were calculated by subtracting the averaged RT to good stimuli from the RT to bad stimuli in each session and plotted in a histogram (Figs. 2f and S2d, e).

Task-related response. To examine task-related neuronal responses, we isolated the activity of a single neuron and counted the numbers of spikes during the T-VRT and V-VRT. To assess the task-related responses, we counted the spikes and compared the spike rates between each targeted test window and the control window: control windows were set to 200–0 ms before the 1st finger-in cue on, and test windows were set to 0–500 ms after each task event started. The analyzed task events were (1) 1st finger-in cue-on, (2) 1st finger-in, (3) Stimulus-on, (4) Stimulus-off, (5) 1st finger-out, (6) 2nd finger-in cue-on, (7) 2nd finger-in, (8) Reward, (9) 2nd finger-in cue-off, (10) 2nd finger-out. To assess statistical significance in task-related responses, we used the Wilcoxon rank-sum test to compare spike counts between the control and test windows in individual trials.

Value-coding activity during the first stimulus presentation. To examine the degree of value discrimination response, we measured the magnitude of each single neuron's activity in response to each tactile or visual stimulus by counting the numbers of spikes within the test windows in individual trials. To investigate the value response triggered by stimulus presentation, we analyzed neural responses during the stimulus presentation (0–500 ms time window after stimulus onset) in both the T-VRT and V-VRT. The firing rates of individual putamen neurons in response to good and bad stimuli were compared using the Wilcoxon rank-sum test to assess the statistical significance of value discrimination. Neurons exhibiting exclusive value discrimination responses in the T-VRT but not the V-VRT were categorized as tactile-selective value neurons. Conversely, those demonstrating value discrimination responses solely in the V-VRT but not the T-VRT were identified as visual-selective value neurons. Neurons showing value discrimination responses in both the T-VRT and V-VRT were classified as bimodal value neurons.

Value-coding activity during the delay period. We unexpectedly observed neurons exhibiting stronger value discrimination responses during the delay period than during the stimulus presentation period. We thus analyzed neural responses during a 500 ms time window following complete finger withdrawal in both the T-VRT and V-VRT. During this time window, the monkeys were presented with only a blank screen to allow us to isolate the value discrimination response from the potential influences of sensory inputs and movements. The same categorization method used for the stimulus neurons was applied to classify the types of value neurons during the delay period.

Some neurons exclusively exhibited a value discrimination response during the delay period or the stimulus presentation period, whereas others showed a more strongly biased response in either period. Neurons were categorized based on their responses in each period using a receiver operating characteristic (ROC) curve calculated with the firing rates of individual neurons to good stimuli versus bad stimuli. A small subset of neurons sustained their value discrimination responses across both periods, leading to the classification of two types of value neurons.

Stimulus and value discrimination index. To determine whether value neurons exhibited stimulus-selective responses, we compared spike numbers for each tactile or visual stimulus in individual trials using the Wilcoxon rank-sum test. For an overall illustration of stimulus-selective responses by value-coding neurons, we measured the magnitudes of responses to each braille pattern and fractal image using the area under the ROC curve. Selectivity responses to tactile stimuli were analyzed with spike counts from both tactile-selective and bimodal value neurons. Selectivity responses to visual stimuli were analyzed using spike counts from both visual-selective and bimodal value neurons. An ROC value of 0.5 indicates identical responses to both stimuli, and values closer to 1 or 0 indicate a stronger response to a specific stimulus (Fig. 3f).

We calculated the value discrimination index using an ROC analysis based on the magnitudes of neural responses to good versus bad stimuli in both the T-VRT and V-VRT (Fig. 4h, i). ROC areas higher and lower than 0.5 indicate positive and negative value-coding, respectively.

Correlation between neural response and finger-in reaction time. To examine the correlation between the responses of each neuron and the second finger-in RT across trials, we initially calculated the number of spikes in the respective value-coding windows. To analyze the correlation between the activity of stimulus value neurons and finger insertion speeds, spike counting was performed in the 0–500 ms window from stimulus onset. To analyze the correlation between the activity of delay value neurons and finger insertion speeds, the analysis included the count of spikes in the 0–500 ms after finger withdrawal. To assess how neural responses and RTs changed around the time of value reversal, spike counts in each bin of two trials were averaged and plotted in graphs (Figs. 5a, b and S9). As each trial within each session was pseudo-randomized, good and bad valued trials are intermingled. Consequently, bin numbers did not reflect consecutive presentations of each valued stimulus but are organized according to the sequence of either good or bad stimulus presentations in which they are presented within each block.

To analyze dynamic changes in correlations, we calculated the number of spikes for both stimulus and delay value neurons in both windows during the stimulus presentation period (0–500 ms from stimulus onset) and the delay period (0–500 ms after finger withdrawal). Correlations between reaction times (RTs) and neural activities were calculated during periods when the neurons encoded value information. Additionally, we calculated correlations between RTs and neural activities during periods when value information was not represented by the neurons to test the possibility that these neurons still guide the speed of RTs regardless of encoding values. Subsequently, we investigated whether the second finger-in RTs in each trial correlated with neural counts in these two distinct windows. A Pearson correlation analysis was conducted to assess statistical significance. Based on the coefficient r values, we use a histogram to represent the number of each type of value-coding neurons exhibiting statistically significant correlations (Figs. 5c and S10).

The correlation coefficient typically ranges between −1 and 1 because each neuron type includes both positive and negative value neurons. Consequently, positive value neurons are negatively correlated with finger RT, whereas negative value neurons are positively correlated with finger RT.

Neural population decoding

Time window. This analysis was conducted to understand how value information is processed in the population responses of putamen value neurons. In the decoding analysis, both groups of value neurons were considered together using the spike data in two time windows: one during the stimulus period aligned to stimulus onset (−200–500 ms) and the other during the delay period aligned to finger withdrawal (0–500 ms). We calculated the spike train of each 25 ms bin within each period and concatenated the spike counts in these two periods for the subsequent decoding analysis.

Neural decoding analysis with simulated neural population. To compare the efficiency of parallel and convergent value processes, we performed a neural decoding analysis with modality-selective value neurons ($n = 118$) and bimodal value neurons ($n = 129$) derived from our original neural data. This analysis used the maximum correlation coefficient classifier method developed by Meyers[60]. The classifier was trained to discriminate values or modalities of stimuli and calculate decoding accuracies. To simulate various ratios of modality-selective and bimodal value neurons, value neurons from each group were randomly sampled. We used several different ratios, with tactile-selective, bimodal, and visual-selective value neurons distributed 1:0:0, 0:0:1 (modality-selective value neurons), 1:0:1 (parallel value process), 1:1:1, 1:2:1, 1:3:1, 1:4:1, and 0:1:0 (convergent value processes). To simulate a larger number of neurons than our original dataset, we performed neuron augmentation by creating synthetic neurons through random sampling of actual neuronal responses in trials from each modality group.

In the decoding analysis, the firing rates of individual neurons were collected using a 25 ms bin width, as described earlier, and normalized by z-score. Trials from each group were randomly divided into training sets and test sets using 10-fold cross-validation. The classifier was trained on data from 9 splits and tested on the 10th split. This cross-validation procedure was repeated 10 times with a different test split each time. The entire process was replicated over 50 resample runs, creating different random populations of training and test splits in each run. The decoding accuracy data were averaged across these resample runs to obtain a single set of results. This entire procedure was iterated more than 50 times, and the results were used as the main dataset for analysis and visualization.

To estimate the statistical significance of the decoding accuracy, a permutation test was performed by shuffling the labels of each trial variable in each neuron. The same decoding procedure was used to generate a null distribution of decoding accuracies. This procedure was repeated 960 times and generated a fully shuffled decoding null distribution for each time bin. We calculated the time points at which the decoding results were greater than all 960 decoding results of the null distribution. The times when the decoding accuracies significantly exceeded the chance level for at least 5 consecutive bins are indicated as solid bars on the bottom of Fig. 6b.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data are available from the corresponding authors on request. Sharing and reuse of data require the expressed written permission of the authors, as well as clearance from the Institutional Review Boards and the Institutional Animal Care and Use Committee of Seoul National University. Source data are provided with this paper.

Code availability

The custom codes are available from the corresponding authors on request. Sharing and reuse of data require the expressed written permission of the authors.

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Author contributions

H.F.K. and S.-H.L. designed and supervised the entire project. S.H.H., D.P., J.W.L., S.-H.L., and H.F.K. performed the behavior, fMRI, and single-unit recording experiments. S.H.H., D.P., and J.W.L. analyzed the data and prepared the figures. S.H.H., D.P., and J.W.L. wrote the first draft, and S.H.H., D.P., J.W.L., S.-H.L., and H.F.K. interpreted data and wrote the final manuscript.

Competing interests

The authors declare no competing interests.

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